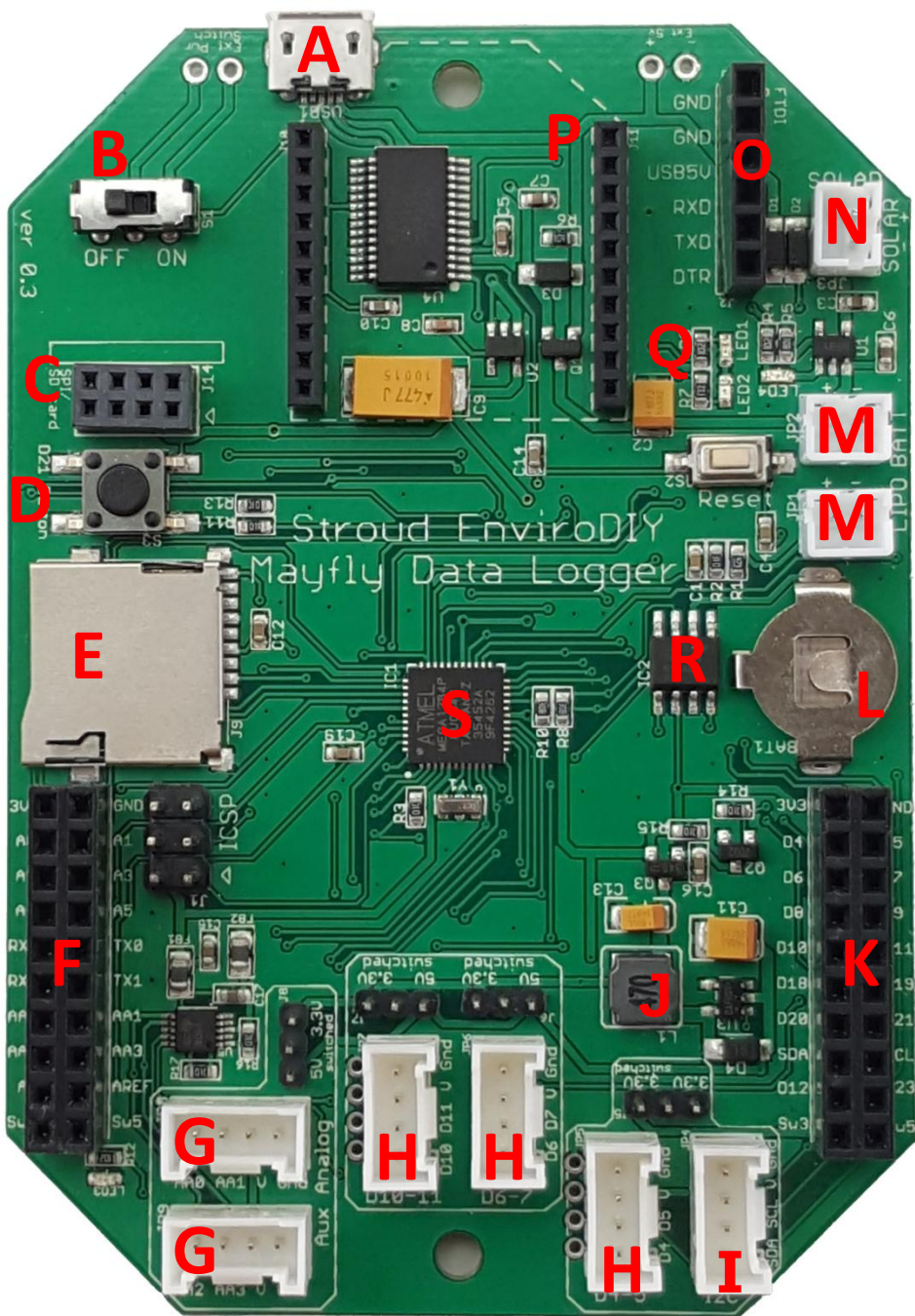




Features of the Mayfly Data Logger

v0.3, 0.4, 0.5, & 0.5b

A	MicroUSB port – connect a standard MicroUSB cable to a computer for programming the Mayfly using the Arduino software
B	Power switch – turns the Mayfly board on and off
C	microSD/SPI connector – socket for vertical microSD memory card adapter board or other SPI devices
D	Pushbutton – connected to pin D21 for user-defined input
E	microSD card socket – socket for storing data on a standard microSD memory card
F	Analog pin header – access to the Mayfly’s power, ground, & analog pins, and also the four Auxiliary 16-bit Analog-to-Digital converter pins
G	Auxiliary ADC Grove connectors – pairs of Auxiliary Analog pins along with ground and power (3.3v or 5V)
H	Digital pin Grove connectors – pairs of digital pins along with ground and power (3.3v or 5v), for connecting sensors and Grove accessories
I	I²C port Grove connector – connection for any devices that use the I ² C protocol
J	5-volt boost converter – generates 5v for powering external sensors
K	Digital pin header – access to the Mayfly’s power, ground, & digital pins
L	Clock battery – socket for CR1220 lithium battery to keep clock chip (R) running when no other power is connected to Mayfly
M	LiPo battery connectors – JST socket for connecting LithiumPolymer (LiPo) rechargeable battery. Additional socket is for providing power to high-current peripheral devices
N	Solar panel connector – JST socket for connecting 6v solar panel for charging the LiPo battery
O	FTDI programing header – alternative port for programming board using an external FTDI adapter instead of using the Mayfly microUSB port
P	Bee module socket – connection port for various telemetry modules that use the Bee footprint (mesh radio, WiFi, cellular)
Q	Red & Green LEDs – LEDs for providing visual feedback, connected to pins D8 (green) and D9 (red)
R	Real-time clock – DS3231 clock module with on-board temperature sensor, retains the date and time after initial programming, requires battery (L)
S	Processor – ATmega1284p microprocessor